

Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.

Team Members	Role/ Responsibility	Email
David SHUMBUSHO	preprocessing; PCA core implementation (eigendecomposition); optimization & benchmarking (Task 3)	d.shumbusho@alustudent.com
	covariance; dynamic component selection (Task 2); visualization, written interpretations & documentation	
Keyla Nyacyesa		k.nyacyesa@alustudent.com

Task Allocation and Tracker					
Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
Finalize prepare Africanized dataset (≥ 7 cols, missing values, non-numeric column)	David SHUMBUSHO	03 Jun 2026	Completed	Yes	African Development Indicators – 57 countries, 20 columns
Step 1: Load data (NumPy only) & handle missing values via mean imputation	David SHUMBUSHO	04 Jun 2026	Completed	Yes	24 missing values imputed; Country/Region set aside
Step 1: Standardize data using $z = (x - \mu) / \sigma$	Keyla Nyacyesa	04 Jun 2026	Completed	Yes	Verified mean ≈ 0 , std ≈ 1 per feature
Step 2: Compute covariance matrix + written explanation (≥ 2 reasons)	Keyla Nyacyesa	05 Jun 2026	Completed	Yes	$\Sigma = Z^T Z / (n-1)$
Steps 4 & 5: Eigendecomposition (eigh) and sort components descending	David SHUMBUSHO	05 Jun 2026	Completed	Yes	Used np.linalg.eigh for symmetric matrix
Task 2: Dynamic component selection by explained variance + scree plot	Keyla Nyacyesa	06 Jun 2026	Completed	Yes	90% threshold $\rightarrow$ 10 of 18 components (92.56%)

Steps 0–6: Projection, reduced-data output & before/after visualization	David SHUMBUSHO	07 Jun 2026	Completed	Yes	Reduced from 18 → 10 features; reconstruction check
Task 2: Optimize implementation (vectorized & SVD) and benchmark	David SHUMBUSHO	08 Jun 2026	Completed	Yes	Vectorized $\approx 28\times$ faster than naive loop
written interpretations, README & final review / submission	Keyla Nyacyesa	10 Jun 2026	Completed	Yes	Interpretations written in own words; combined PDF prepared

Meeting Attendance Log					
Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps
03 Jun 2026	Project kickoff & planning	David SHUMBUSHO	David SHUMBUSHO, Keyla Nyacyesa	Selected the African Development Indicators dataset; agreed on the from-scratch PCA pipeline and split roles between the two members	source & clean data and set up notebook; Keyla: prepare standardization & covariance
05 Jun 2026	Core PCA progress check	Keyla Nyacyesa	David SHUMBUSHO, Keyla Nyacyesa	Reviewed loading, missing-value imputation, standardization, covariance and eigendecomposition; confirmed all cell outputs render correctly	Implement dynamic component selection (Task 2) and the before/after visualization
08 Jun 2026	Task 2 & Task 3 review	David SHUMBUSHO	David SHUMBUSHO, Keyla Nyacyesa	Reviewed explained-variance selection (90% → 10 components) and benchmark results (vectorized vs SVD vs naive)	Write interpretations in our own words; finalize README and projection plots

10 Jun 2026	Final review & submission	Keyla Nyacyesa	David SHUMBUSHO, Keyla Nyacyesa	Proofread all interpretations, verified outputs are visible, generated the combined PDF and completed this task sheet	notebook, dataset and PDFs to the GitHub repo and submit the link
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